MAH Model Rebuild: Main-Text Model Candidate

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Abstract

This candidate is the restrained main-text split of the approved Phase 15 derivation draft. The MAH reform makes retained entrusted manufacturing legally available by changing one institutional wedge. Developers choose one common project-advancement intensity before project characteristics and organizational routes are realized. Route choice is deterministic, and the price of qualified manufacturing services is endogenous. The baseline contains neither patent-generating research nor a fixed ranking between original and incremental project responses.

3 Model

3.1 Environment and timing

There is a continuum of developers. Developer i has predetermined research capability $a_i > 0$ and internal manufacturing capability $k_i > 0$, distributed according to $H(a, k)$. A viable project that reaches commercialization-relevant planning draws value $q > 0$ and manufacturing requirement $m > 0$ from the exogenous distribution $F(q, m)$. For empirical decompositions only,

$$F(q, m) = \sum_{g \in \{O, \mathrm{Inc}\}} \rho_g F_g(q, m), \qquad \rho_g \geq 0, \quad \sum_g \rho_g = 1.$$

The class label g creates no class-specific control.

The institutional regime is $M \in \{0, 1\}$. The pre-MAH regime sets the barrier to retained entrusted manufacturing at infinity; the MAH regime makes it finite:

$$\tau_E(0) = +\infty, \qquad \tau_E(1) = \bar{\tau}_E < +\infty, \qquad \bar{\tau}_E \geq 0. \tag{1}$$

The available routes are internal manufacturing I, retained entrusted manufacturing E, non-retained transfer T, and abandonment or delay A. Under E, the developer remains the authorization holder and a qualified external producer manufactures. Thus E is not transfer.

Before project characteristics and routes are realized, developer i chooses one common $x_i \geq 0$, interpreted as project-development / advancement intensity. It advances viable drug projects toward commercialization-relevant route planning and excludes basic research, compound discovery, patent-generating effort, and patent applications. The expected measure of viable projects reaching route planning is

$$\lambda_i^{\mathrm{plan}} = a_i x_i. \tag{2}$$

Timing is as follows. First, M is observed and developers anticipate the market-consistent qualified-service price. Second, they choose x_i. Third, planning-stage projects draw (q, m) and choose a route. Fourth, downstream realization occurs with exogenous probability $s(q)$. Finally, aggregate capacity demand and supply must validate the anticipated service price. The last step is an equilibrium consistency condition, not a late unanticipated shock. Upstream research and patent generation may help determine the predetermined a_i, but they are outside the baseline decision problem.

3.2 Commercialization technologies

Conditional on successful commercialization, residual demand is $y(p;q) = Aqp^{-\varepsilon}$, where $A > 0$ and $\varepsilon > 1$. Product-price optimization gives

$$p^*(c) = \frac{\varepsilon}{\varepsilon - 1}c$$

and the gross operating present value

$$R(q,c) = \frac{Aq}{1-\beta\varphi}\frac{(\varepsilon-1)^{\varepsilon-1}}{\varepsilon^{\varepsilon}}c^{1-\varepsilon}, \qquad \beta \in (0,1), \quad \varphi \in [0,1). \tag{3}$$

The appendix derives the FOC, SOC, and accounting. In particular, $R_q > 0$ and $R_c < 0$.

Internal production uses marginal cost $c_I(m, k_i)$, with $c_{I,m} > 0$ and $c_{I,k} < 0$, and setup cost $F_I(m, k_i)$, with $F_{I,m} > 0$ and $F_{I,k} < 0$ on the feasible domain. Internal production is unavailable when $k_i < \underline{k}(m)$, represented by $F_I(m, k_i) = +\infty$. On the feasible domain its route value is

$$W_i^I(q,m) = s(q)R(q, c_I(m,k_i)) - F_I(m,k_i). \tag{4}$$

Below the feasibility threshold define $W_i^I = -\infty$ directly; c_I need not be defined there. Cost derivatives apply only to the feasible interior.

Entrusted production uses external marginal cost $c_E(m)$, requires $b(m) > 0$ units of qualified capacity with $b'(m) > 0$, incurs setup cost $F_E(m)$, and leaves the holder with burden $\mu_E \geq 0$. At the qualified-capacity price p_m,

$$\begin{aligned} W_i^E(q,m;M,p_m) = s(q)R(q,c_E(m)) - F_E(m) - p_m b(m) \\ - \mu_E - \tau_E(M). \end{aligned} \tag{5}$$

Both marginal costs are strictly positive on their domains; setup costs are nonnegative when finite. The capacity payment $p_m b(m)$ is excluded from c_E and R. Each monetary cost enters once. The transfer and abandonment values are

$$W^T(q,m) = T(q,m), \qquad W^A = 0. \tag{6}$$

3.3 Organization and project advancement

At a fixed p_m, the project chooses deterministically:

$$\begin{aligned} W_i(q,m;M,p_m) &= \max\{W_i^I, W_i^E, W^T, W^A\}, \\ r_i^*(q,m;M,p_m) &\in \arg\max_{r\in\{I,E,T,A\}} W_i^r. \end{aligned} \tag{7}$$

Here $W_i^T = W^T$ and $W_i^A = W^A$. We assume measurable values and zero probability of maximizing ties at each candidate price and regime for almost every developer; continuity of heterogeneity alone is not sufficient. A measurable tie rule completes the definition on null sets.

The internal-minus-entrusted gap is

$$\begin{aligned} \Delta_{IE}(k_i) = s(q)[R(q,c_I(m,k_i)) - R(q,c_E(m))] - F_I(m,k_i) \\ + F_E(m) + p_m b(m) + \mu_E + \tau_E(M). \end{aligned} \tag{8}$$

Holding (q,m,M,p_m) fixed with finite τ_E, on the internally feasible interior,

$$\Delta_{IE,k} = s(q)R_c(q,c_I)\,c_{I,k} - F_{I,k} > 0.$$

When the continuous gap has a negative lower-end limit and a positive upper-end limit, a unique conditional cutoff $k^*(q,m;p_m,M)$ satisfies $\Delta_{IE}(k^*)=0$. Conditional on I and E dominating T and A, developers below the cutoff select E, while those above it select I. Transfer and abandonment remain valid outside options.

Expected optimized value per planning-stage project is

$$\Omega_i(M,p_m) = \int W_i(q,m;M,p_m)\,dF(q,m). \tag{9}$$

Developer i solves

$$\max_{x_i\ge 0}\left\{\beta a_i x_i \Omega_i(M,p_m) - \frac{\kappa}{1+\nu}x_i^{1+\nu}\right\}, \qquad \kappa>0, \quad \nu>0. \tag{10}$$

Strict concavity gives the unique solution

$$x_i^*(M,p_m) = \left[\frac{\beta a_i}{\kappa}\Omega_i(M,p_m)\right]^{1/\nu}, \tag{11}$$

including $x_i^*=0$ when $\Omega_i=0$. The binary reform affects x_i^* only through Ω_i. At a fixed p_m, its effect is strict only if E improves on the pre-MAH maximum on a positive-probability project set.

3.4 Qualified manufacturing-service equilibrium

Qualified supplier j, with efficiency z_j, chooses capacity $s_j \ge 0$:

$$\max_{s_j\ge 0}\{p_m s_j - \Psi(s_j;z_j)\}. \tag{12}$$

Assume $\Psi(0;z)=\Psi_s(0;z)=0$, $\Psi_{ss}>0$, and marginal cost diverges with capacity. At a positive price, capacity is uniquely determined by $p_m=\Psi_s(s_j^*;z_j)$, and aggregate supply is

$$S_m(p_m) = \int s_j^*(p_m,z)\,dH_C(z).$$

Expected entrusted capacity per planning-stage project is

$$\chi_i^E(p_m;M) = \int b(m)\mathbf{1}\{r_i^*(q,m;M,p_m)=E\}\,dF(q,m). \tag{13}$$

Study-related and total demand are

$$\begin{aligned} D_m^{\mathrm{MAH}}(p_m;M) &= \int a_i x_i^*(M,p_m)\chi_i^E(p_m;M)\,dH(a,k),\\ D_m(p_m;M) &= D_m^B(p_m) + D_m^{\mathrm{MAH}}(p_m;M). \end{aligned} \tag{14}$$

The demand expression includes both feedbacks: price changes expected value and hence advancement, and it changes deterministic route selection. The no-maximizing-ties condition, continuity of finite route values in price, and the integrable value/capacity envelopes stated in the appendix make aggregate demand continuous even though individual indicators can jump. Supplier responses are integrable on compact price intervals. Background demand is finite, continuous, weakly decreasing, positive at zero, and tends to zero at infinity.

The qualified-capacity market clears at the unique price

$$D_m(p_m^*; M) = S_m(p_m^*). \tag{15}$$

Supply is strictly increasing and demand weakly decreasing. Positive background demand gives excess demand at zero, while entrusted and background demand vanish and supply diverges at high prices. These conditions give existence and uniqueness. If post-MAH study demand is positive at the old price, $p_m^*(1) > p_m^*(0)$: CMO scarcity attenuates the reform rather than amplifying it through a direct price reduction.

The baseline partial equilibrium contains exactly

$$\left\{p_m^*,\ x_i^*,\ r_i^*(q,m),\ s_j^*\right\}_{i,j}. \tag{16}$$

No entry, labor, capital, product-market aggregate, or welfare closure is added.

3.5 Main predictions

Organizational sorting. Conditional on I and E dominating the outside options, low internal manufacturing capability selects entrusted production and high capability selects internal production. Lower entrusted-route friction expands the entrusted region; a higher fixed CMO price contracts it.

MAH-relevant projects and advancement. The reform has a zero effect whenever the newly feasible E value does not exceed the best old route. Otherwise it raises expected project value and, at fixed service price, raises common project advancement. Holding the initial surplus fixed, the response increases with the value gain. Across developers, higher a_i scales a given surplus response upward; a larger response at lower k_i additionally uses $\nu \geq 1$ and the feasibility-boundary continuity conditions in the appendix. No general capability ranking is imposed for $0 < \nu < 1$. MAH does not change a_i, project quality, or the advancement-cost function.

CMO equilibrium and scarcity. The qualified-capacity price is unique. Reform-induced entrusted demand can raise this price, weakly reducing the direct fixed-price gains in project value and advancement. Fixed-price comparisons and equilibrium-price comparisons are therefore distinct.

Planning, organization, and observed outcomes. At equilibrium,

$$\Lambda_i^{\text{plan}} = a_i x_i^*,$$

while expected realized retained outcomes are

$$Y_i^{\text{ret}} = a_i x_i^* \int s(q)\, [\mathbf{1}\{r_i^* = I\} + \mathbf{1}\{r_i^* = E\}]\, dF(q,m). \tag{17}$$

Thus observed outcomes can change through advancement or route composition. Holder–producer separation is observed only after route assignment and cannot cause the earlier x_i choice.

Novelty composition. For $g \in \{O, \text{Inc}\}$, outcomes use the same x_i^* and the mixture $\rho_g F_g$, with the same realization kernel $s(q)$. A zero class-specific value gain alone does not imply a zero outcome effect: the common advancement response also scales that class's existing retained outcomes. The baseline permits either class to have a zero or larger response and imposes no original-versus-incremental ranking. Patent applications are not a baseline endogenous outcome.